

2026 A-level

Q.7

(a) - *Sector Area Equation*

$$\begin{aligned}
 R_1 &= 2R_2 \\
 \pi r^2 \frac{\theta}{2\pi} - \frac{1}{2} r^2 \sin \theta &= 2 \left[\pi r^2 \frac{\pi - \theta}{2\pi} - \frac{1}{2} r^2 \sin(\pi - \theta) \right] \\
 \frac{\theta}{2} - \frac{1}{2} \sin \theta &= \pi - \theta - \frac{1}{2} \sin \theta \\
 \frac{\sin \theta}{2} &= \pi - \frac{3\theta}{2}
 \end{aligned}$$

Q.8

(a) **Integral:** $\int_0^3 x e^{2x} dx$

(b)

$$\begin{aligned}
 \int_0^3 x e^{2x} dx &= \left[\frac{x e^{2x}}{2} \right]_0^3 - \int_0^3 \frac{e^{2x}}{2} dx \\
 &= \frac{3e^6}{2} - \left(\frac{e^6}{4} - \frac{1}{4} \right) \\
 &= e^6 + \frac{1}{2}
 \end{aligned}$$

Q.9

(a) - *Trigonometric Identities*

$$\begin{aligned}
 (\cos \theta + \sin \theta) \left(\frac{1}{\sin \theta} - \frac{1}{\cos \theta} \right) &= (\cos \theta + \sin \theta) \frac{(\cos \theta - \sin \theta)}{\sin \theta \cos \theta} \\
 &= \frac{\cos^2 \theta - \sin^2 \theta}{\sin \theta \cos \theta} \\
 &= \frac{1 - \tan^2 \theta}{\tan \theta} \\
 &= 2 \times \frac{1 - \tan^2 \theta}{2 \tan \theta} \\
 &= 2 \cot 2\theta
 \end{aligned}$$

(b)

$$\begin{aligned}
 2 \cot 2\theta &= \frac{4}{5 \sin^2 2\theta} \\
 2 \sin 2\theta \cot 2\theta &= \frac{4}{5} \\
 \sin 4\theta &= \frac{4}{5} \\
 4\theta &= \dots\dots\dots - 360^\circ < x < 360^\circ
 \end{aligned}$$

Q.10**(a)** - Differentiation, slope at $x = 2$

$$\begin{aligned}
 y &= \frac{7x}{\sqrt{3x^2 + 4}} \\
 \frac{dy}{dx} &= \frac{7\sqrt{3x^2 + 4} - 7x \cdot \frac{1}{2}(3x^2 + 4)^{-\frac{1}{2}}(6x)}{3x^2 + 4} \\
 &= \frac{7}{\sqrt{3x^2 + 4}} - \frac{21x^2}{(3x^2 + 4)^{\frac{3}{2}}} \\
 \frac{dy}{dx} \Big|_{x=2} &= \frac{7}{4} - \frac{84}{64} \\
 &= \frac{7}{16}
 \end{aligned}$$

When $x = 2, y = \frac{14}{4} = \frac{7}{2}$.

$$y - \frac{7}{2} = -\frac{16}{7}(x - 2)$$

$$14y - 49 = -32x + 64$$

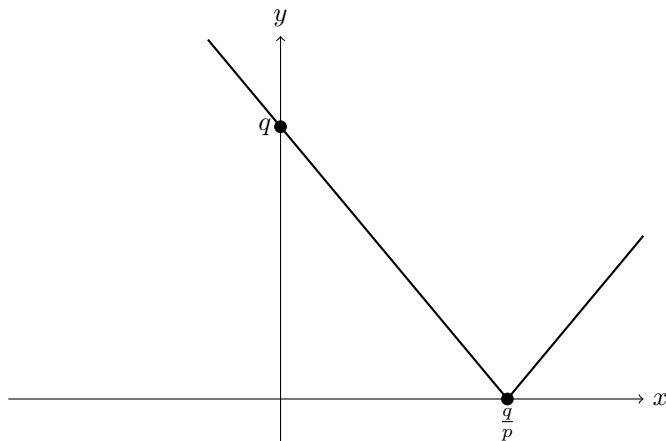
$$32x + 14y - 113 = 0$$

(b) When $x = 0, y = \frac{113}{14}$.**(c)** $d(3x^2 + 4) = 6x \, dx$

$$\begin{aligned}
 \int_0^2 \left(\frac{113 - 32x}{14} - \frac{7x}{\sqrt{3x^2 + 4}} \right) dx &= \left[\frac{113}{14}x - \frac{32x^2}{28} \right]_0^2 - \int_0^2 \frac{7}{6}(3x^2 + 4)^{-\frac{1}{2}} d(3x^2 + 4) \\
 &= \frac{113}{7} - \frac{32}{7} - \left[\frac{7}{6} \left(\frac{1}{2} \right) (3x^2 + 4)^{\frac{1}{2}} \right]_0^2 \\
 &= \frac{81}{7} - \frac{7}{12}(4 - 2) \\
 &= \frac{81}{7} - \frac{7}{6} \\
 &= \frac{437}{42}
 \end{aligned}$$

Q.12**(a)** - Absolute function

$$y = \begin{cases} px - q, & x \geq \frac{q}{p} \\ q - px, & x < \frac{q}{p} \end{cases}$$



(b) Consider $y = \frac{k}{x}$ and $y = q - px$.

$$\begin{aligned}q - px &= \frac{k}{x} \\qx - px^2 &= k \\px^2 - qx + k &= 0\end{aligned}$$

$$\begin{aligned}\Delta &> 0 \\q^2 - 4pk &> 0 \\k &< \frac{q^2}{4p}\end{aligned}$$

Q.13

(a) - *Geometric Sequence*

$$\sin \theta, \sqrt{2} \cos \theta, \sqrt{3} \cot \theta \quad 0 < \theta < \frac{\pi}{2}$$

$$\begin{aligned}\frac{\sqrt{2} \cos \theta}{\sin \theta} &= \frac{\sqrt{3} \cot \theta}{\sqrt{2} \cos \theta} \\2 \cos \theta &= \sqrt{3} \\\cos \theta &= \frac{\sqrt{3}}{2}\end{aligned}$$

$$\begin{aligned}r &= \frac{\sqrt{2} \cos \theta}{\sin \theta} \\&= \frac{\sqrt{2} \cdot \frac{\sqrt{3}}{2}}{\frac{1}{2}} \\&= \sqrt{6}\end{aligned}$$

(b)

$$\begin{aligned}\frac{\sin \theta}{(\sqrt{6})^2} + \frac{\sin \theta}{\sqrt{6}} + \sin \theta &= \frac{1}{6} \times \frac{1}{2} + \frac{1}{\sqrt{6}} \times \frac{1}{2} + \frac{1}{2} \\&= \frac{7}{12} + \frac{1}{2\sqrt{6}} \\&= \frac{7}{12} + \frac{\sqrt{6}}{12} \\&= \frac{7 + \sqrt{6}}{12}\end{aligned}$$

Q.14**(a)** - Integrate a differential equation

$$\frac{dx}{dt} = x(A - t)$$

$$\frac{dx}{x} = (A - t)dt$$

$$\ln x = At - \frac{t^2}{2} + C$$

When $t = 0$, $x = 0.3$. $\implies \ln 0.3 = C$

When $t = 4$, $x = 21$.

$$\ln 21 = 4A - 8 + \ln 0.3$$

$$A = \frac{8 + \ln 21 - \ln 0.3}{4}$$

$$A = 2 + \frac{\ln 70}{4}$$

$$\ln x = \left(2 + \frac{\ln 70}{4}\right)t - \frac{t^2}{2} + \ln 0.3$$

$$x = e^{(2 + \frac{\ln 70}{4})t - \frac{t^2}{2} + \ln 0.3}$$

(b) - Maximum: slope = 0Note that $\frac{dx}{dt} = 0$ when $x(A - t) = 0$.

$$\left(2 + \frac{\ln 70}{4} - t\right) e^{(2 + \frac{\ln 70}{4})t - \frac{t^2}{2} + \ln 0.3} = 0$$

$$2 + \frac{\ln 70}{4} - t = 0$$

$$t = 2 + \frac{\ln 70}{4}$$

$$= 3.06212\dots$$

First derivative:

$$\frac{dx}{dt} = x(A - t)$$

$$= \left(2 + \frac{\ln 70}{4} - t\right) e^{(2 + \frac{\ln 70}{4})t - \frac{t^2}{2} + \ln 0.3}$$

Second derivative:

$$\frac{d^2x}{dt^2} = -e^{(2 + \frac{\ln 70}{4})t - \frac{t^2}{2} + \ln 0.3} + \left(2 + \frac{\ln 70}{4} - t\right) \left[\left(2 + \frac{\ln 70}{4}\right) - t\right] e^{(2 + \frac{\ln 70}{4})t - \frac{t^2}{2} + \ln 0.3}$$

Check that when $t = 2 + \frac{\ln 70}{4}$,

$$\left.\frac{d^2x}{dt^2}\right|_{t=2+\frac{\ln 70}{4}} = -e^{[\dots]} < 0 \implies \text{Maximum}$$

Q.15 - *Discrete Maths*

$$\begin{aligned}(2m+1)^2 + (2n+1)^2 &= c^2 \\ 4m^2 + 4m + 1 + 4n^2 + 4n + 1 &= c^2 \\ 2[2(m^2 + n^2) + 2(m+n) + 1] &= c^2\end{aligned}$$

Since 2 is a factor of c^2 , i.e., c^2 is even.

$\therefore$ c is even.

Let $k = 2p$, for some $p \in \mathbb{N}$.

$$\begin{aligned}2[2(m^2 + n^2) + 2(m+n) + 1] &= (2p)^2 = 4p^2 \\ 2(m^2 + n^2) + 2(m+n) + 1 &= 2p^2\end{aligned}$$

However, $2(m^2 + n^2) + 2(m+n) + 1$ cannot be even.

Contradiction!!!